Healthcare Process Modelling: Leveraging Domain Expertise to Identify EHR Data Proxies for Behavioral Data

Learning Objectives

By the end of this presentation, viewers will be able to:

- Define Healthcare Process Modelling
- Understand the motivation behind the CONCERN study
- Understand the value of EHR Metadata when it comes to EWS
- Appreciate the importance of domain expertise
- Evaluate the design and functionality of the CONCERN model

Motivation

-Many preventable in-patient deaths

-Nurses have different perspectives from physicians

-Suboptimal interactions and communication

- Higher proportion of cardiac arrest patient had 5+ comments from RN

-Nurses observe subtle changes in behavior that could be interpreted from the EHR

CONCERN Study

- Determine notification threshold for at risk patients by performing analytics of nursing documentation patterns

- Develop an app to do so

    - Had to harmonize data from two sources (NYPres & BWH)

    -Used i2b2 interface

- Study the effectiveness/impact of the app

CONCERN Study cont.

-Used flow sheet metadata from the EHR

-Frequency of documentation

-Free text comments

-Increased Surveillance

All proxies for clinical concern

Clinical Support Decision Making App

Classified patient data using:

- Vital sign comment frequency
- Medication administration patterns (PRN’s in particular)
- Notes w/ special keywords

Patients who experienced cardiac arrest had had significantly more documentation activity in the 48H leading up to the event

Challenges

Data Harmonization

- Two different EHR systems (Epic vs. Allscripts)
- Different labelings, dirty data
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CONCERN Predictive Model Development

- **Multinomial Gradient Boosted Machine (GBM)**
  - Classification: Patient discharge, remain in hospital, or has a negative event
  - Trained on 70% of dataset, 30% used for 10-fold cross-validation
- Factors: Vital Sign Comment Frequency, Vital Sign Entry Frequency, MAR frequency, Nursing Note Frequency, Nursing Note Content, Demographics
- PRN (pro re nata, “as needed”) medications [within MAR frequency]
  - Predictive or protective of patient deterioration
- Numerical: Comment Frequencies + Note Content with Continuous Time Scaling
- Categorical: Clinical workflow factors (eg. demographics, location, etc.)
- Built Time Independent Clinical Deterioration Model from numerical + categorical

NLP Model

- Identified clinical care terms from nursing subject matter experts
- Asked them to rate concepts to inform natural language processing (NLP)
- LDA Topic modelling for rapid response patients for clinical meaning
- Survival analysis for rapid response events (association)

Model Evaluation

- Evaluated model using accuracy, precision, recall, logloss and AUC metrics
  - Achieved 97% accuracy in both ICU and ACU settings
- Compared CONCERN model with other early warning scores
- Higher risk score implied greater risk compared to NEWS and MEWS high risk scores
- Compared log likelihood ratio of observing CONCERN high risk score 42 hours earlier than a MEWS or NEWS high risk score
- Demographics used in model building to reduce potential racial biases, which was not considered in MEWS or NEWS high risk score
- Benefits: Used both data types and temporality; based on documentation frequency, derived from expert domain knowledge, triggered alarm 42 hrs earlier
- Confirmed that how, when, and why a clinical observation is documented impacts information signals

Using CONCERN Model in Clinical Practice

- Problem: Concern doesn’t translate to patient care
- User Centered Design & Simulation Testing of SmartApp - test data presentation & impact on patient awareness
- Designed a multi-site clinical trial testing SmartApp with multiple time-series intervention
- Implementation of dashboard into EPIC care system and training

Summary of Major Points

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